Selection or adaptation under neoadjuvant PD-1 blockade

Bing Han · independent computational project · public data only · October 2026

THE QUESTION

When a tumour looks different after treatment, two very different things can produce that difference. **Selection**: the cell populations were already present and their proportions changed, because treatment removed some and spared others. **Adaptation**: the same cell states persist and have changed what they express. Bulk measurements cannot separate them, and most single-cell differential expression is reported in a way that cannot either.

DATA

GSE236581 — 22 colorectal cancer patients sampled before and after neoadjuvant anti-PD-1; 975,275 cells, 1.31 billion non-zero matrix entries, 212,862 tumour cells at paired timepoints. Held back as a validation cohort: GSE205506, 19 patients, independent group. Inclusion criteria and the full screening of 453 GEO series were fixed before any dataset was opened.

FINDING 1 — THE CHANGE IS MOSTLY WITHIN CELL STATES

Tumour epithelial cells fall from 37,804 to 22,416 while stromal cells double and T, B and myeloid cells rise. Despite that visible shift, the median per-patient composition share is **0.38** (IQR 0.33–0.48; 5 of 22 patients above 0.5; signed-rank against 0.5 $p = 0.0011$). About three fifths of the absolute gene-level change happens inside cell states. Pre-registered to be invalid if two annotation granularities disagreed by more than 0.20; they differ by **0.0245**.

FINDING 2 — A CELL-LEVEL TEST ON DATA CONTAINING NOTHING

Pre-treatment cells only, patients split at random 200 times, so no effect exists by construction. 15,716 genes tested; a calibrated test should call about 786 at $p<0.05$ and none after correction.

Cells per patient	Cell-level, Bonferroni	Patient-level, Bonferroni	Replication cohort, cell-level
100	165	0	153
300	951	0	1,023
1,000	**3,324**	**0**	**4,394**

The false-positive count tracks how many cells were pooled, not any signal: more cells buy more confidence about differences between patients that predate the treatment. Reproduced on the same curve in an independent cohort.

METHOD COMMITMENTS

- Pre-registered before the counts matrix was downloaded: timepoints, floors, statistic, inference, and three conditions that would make the result uninterpretable.
- The patient is the unit of evidence; every inference is across 22 patients.
- The decomposition is an exact identity, asserted on every call, and refuses log input.
- No integration, no re-clustering, no trajectory inference — each excluded with its reason recorded.
- 35 unit tests; the matrix never lands on disk.

WHAT IT DOES NOT CLAIM

- **Not replicated.** The validation cohort was opened once and the comparison proved impossible there — no cell-type annotation, and the pre-registration forbids deriving one. Recorded rather than worked around; the composition finding is permanently single-cohort.
- **Nothing about response.** Responder labels are not in the deposited data and were not inferred.
- **The labels are the authors'.** Agreement across granularities is within one annotation.
- **The audit's patient-level column is conservative**, because the exact test ignores ties — which flatters the comparison, so it is stated.

I designed the study and the statistical approach and interpreted the results; implementation was AI-assisted. Pre-registration, code, tests and the full record of claims refuted by my own controls are available on request.